

Intelligent Control of a Quadrotor Helicopter

ENEL 667 Intelligent Control

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MODEL

Conventional Helicopter

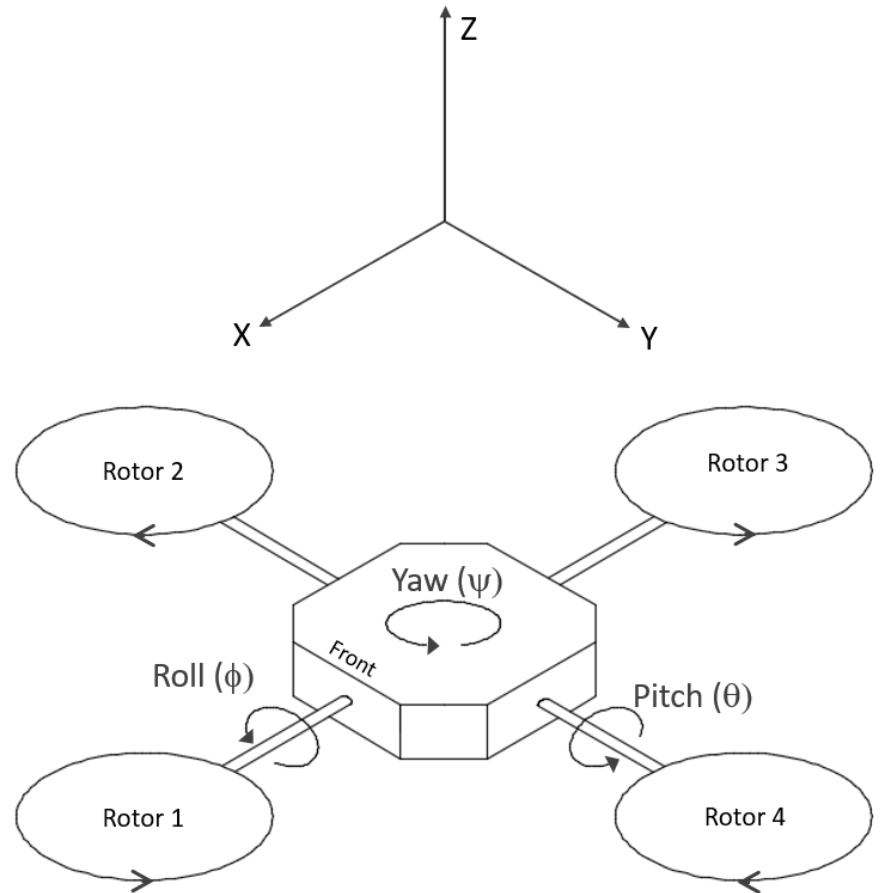


Quadrotor Helicopter



Source: Traxxas, 2015

Quadrotor Coordinate System and Sign Conventions



DYNAMICS EQUATIONS

$$\ddot{z} = -g \cos\phi \cos\theta + \frac{1}{m} F_z$$

$$\ddot{\phi} = \dot{\theta}\dot{\psi} \frac{J_y - J_z}{J_x} - \frac{J_r}{J_x} \dot{\theta}\Omega_r + \frac{1}{J_x} T_\phi$$

$$\ddot{\theta} = \dot{\phi}\dot{\psi} \frac{J_z - J_x}{J_y} + \frac{J_r}{J_y} \dot{\phi}\Omega_r + \frac{1}{J_y} T_\theta$$

$$\ddot{\psi} = \dot{\phi}\dot{\theta} \frac{J_x - J_y}{J_z} + \frac{1}{J_z} T_\psi$$

$$F_z = b (\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2)$$

$$T_\phi = L b (\Omega_4^2 - \Omega_2^2)$$

$$T_\theta = L b (\Omega_3^2 - \Omega_1^2)$$

$$T_\psi = d (\Omega_1^2 + \Omega_3^2 - \Omega_2^2 - \Omega_4^2)$$



$$\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} = \begin{bmatrix} 0.25 & 0 & -0.5 & 0.25 \\ 0.25 & -0.5 & 0 & -0.25 \\ 0.25 & 0 & 0.5 & 0.25 \\ 0.25 & 0.5 & 0 & -0.25 \end{bmatrix} \begin{bmatrix} F_z \\ T_\phi \\ T_\theta \\ T_\psi \end{bmatrix}$$

STATE SPACE SYSTEM

Auxiliary states

$$\begin{aligned}x_1 &= \Lambda z + \dot{z} \\x_2 &= \Lambda \phi + \dot{\phi} \\x_3 &= \Lambda \theta + \dot{\theta} \\x_4 &= \Lambda \psi + \dot{\psi}\end{aligned}$$

Auxiliary state error for trajectory

$$z_3 = x_3 - x_{3\,des} = (\Lambda \theta + \dot{\theta}) - (\Lambda \theta_{des} + \dot{\theta}_{des})$$

State space representation

$$\dot{x} = f(x) + B u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\tilde{x}}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} \Lambda \dot{z} - g \cos \phi \cos \theta \\ \Lambda \dot{\phi} + \dot{\theta} \psi a_1 + \dot{\theta} \Omega_r a_2 \\ \Lambda (\dot{\theta} - \dot{\theta}_{des}) + \dot{\phi} \psi a_3 + \dot{\phi} \Omega_r a_4 - \ddot{\theta}_{des} \\ \Lambda \dot{\psi} + \dot{\phi} \dot{\theta} a_5 \end{bmatrix} + \begin{bmatrix} 1/m & 0 & 0 & 0 \\ 0 & 1/J_x & 0 & 0 \\ 0 & 0 & 1/J_y & 0 \\ 0 & 0 & 0 & 1/J_z \end{bmatrix} \begin{bmatrix} F_z \\ T_\phi \\ T_\theta \\ T_\psi \end{bmatrix}$$

OFFLINE TRAINING

CONTROL LAW AND STABILITY

Lyapunov function candidate, derivative

$$V = \frac{1}{2} x^T B^{-1} x \qquad \dot{V} = x^T (B^{-1} f(x) + u)$$

Neural network approximation

$$B^{-1}f(x) = \begin{bmatrix} m (\Lambda \dot{z} - g \cos \phi \cos \theta) \\ J_x (\Lambda \dot{\phi} + \dot{\theta} \dot{\psi} a_1 + \dot{\theta} \Omega_r a_2) \\ J_y (\Lambda (\dot{\theta} - \dot{\theta}_{des}) + \dot{\phi} \dot{\psi} a_3 + \dot{\phi} \Omega_r a_4 - \ddot{\theta}_{des}) \\ J_z (\Lambda \dot{\psi} + \dot{\phi} \dot{\theta} a_5) \end{bmatrix}$$

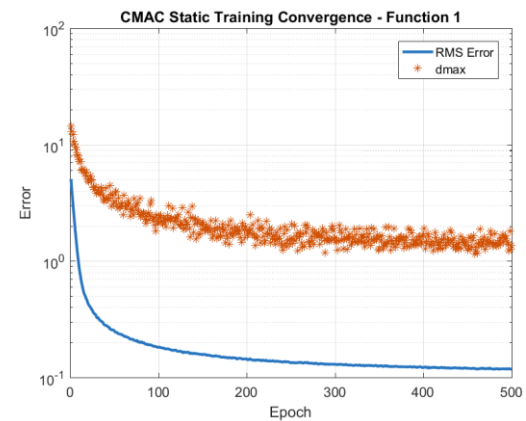
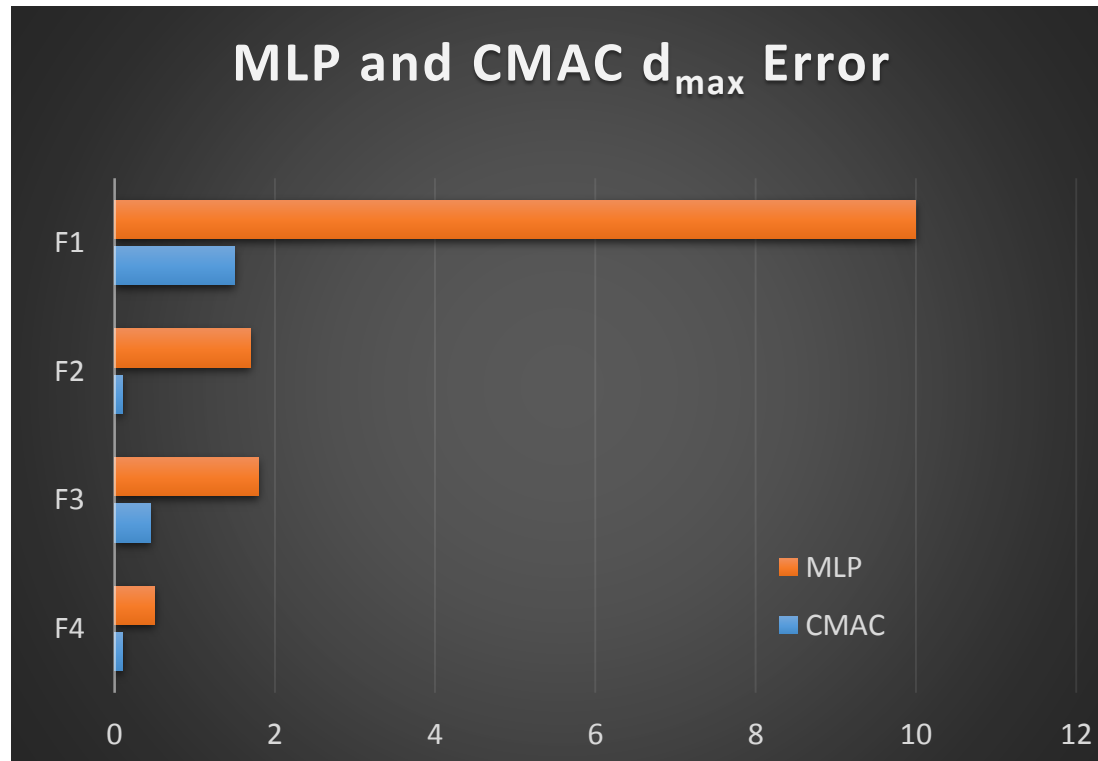
Control law

$$u = -\phi(x) w - K x$$

Lyapunov Stability – x is uniformly ultimately bounded (UUB)

$$\|x\| > \frac{\|d_{max}\|}{K_{min}} \quad \text{when} \quad \dot{V} < 0$$

OFFLINE TRAINING

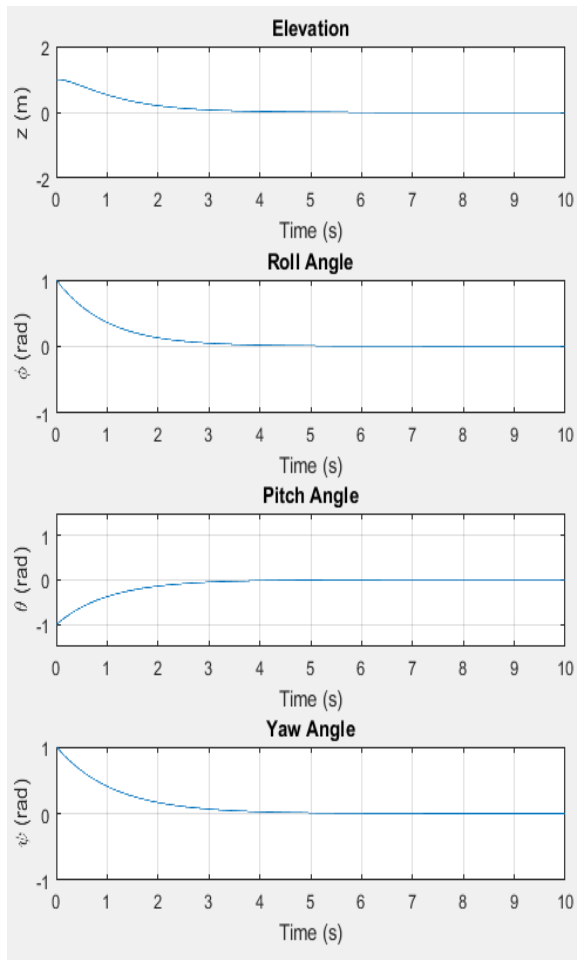


CLOSED LOOP CONTROL

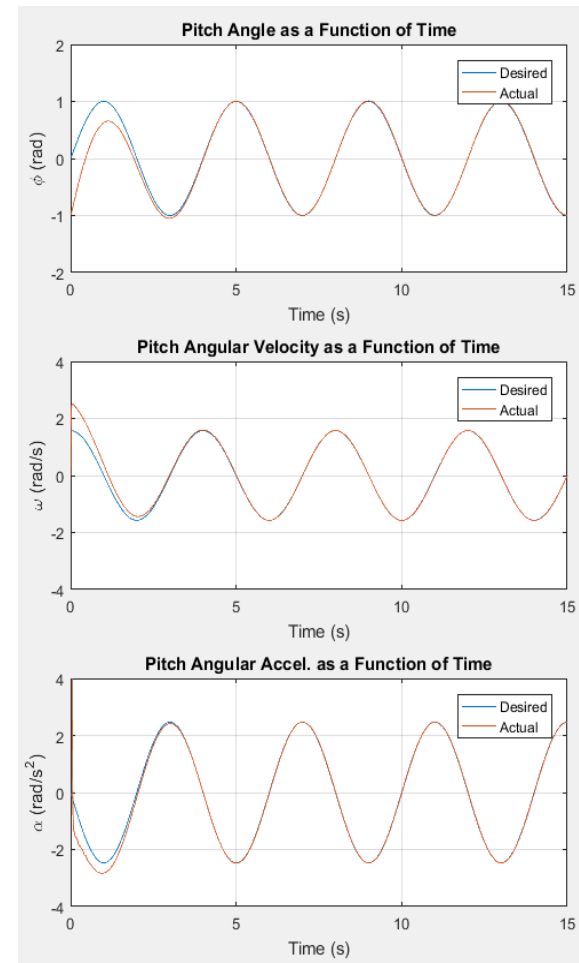
OFFLINE TRAINING

Initial elevation = 1m; initial angles 57°

No trajectory



Sinusoidal trajectory; $\tau=4s$



ONLINE TRAINING

CONTROL AND ROBUST WEIGHTS UPDATE LAWS

Lyapunov function candidate, derivative

$$V = \frac{1}{2} x^T B^{-1} x + \frac{1}{2\beta} \tilde{w}^T \tilde{w}$$

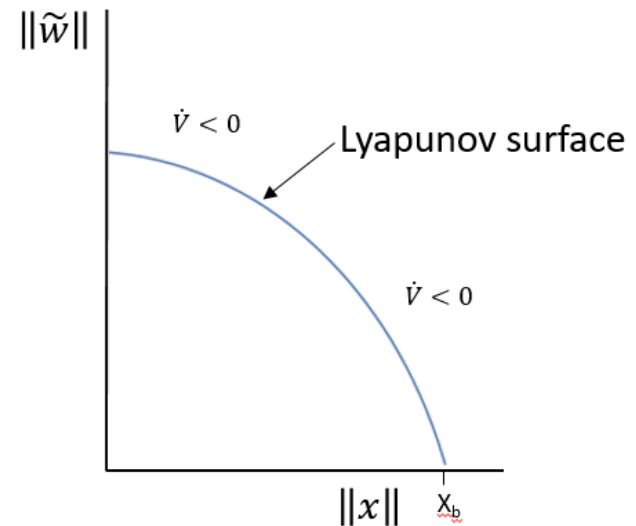
$$\dot{V} = x^T (B^{-1} f(x) + u) - \frac{1}{\beta} \tilde{w}^T \dot{\hat{w}}$$

Control law

$$u = -\Gamma(x) w - K x$$

E-Modification Method

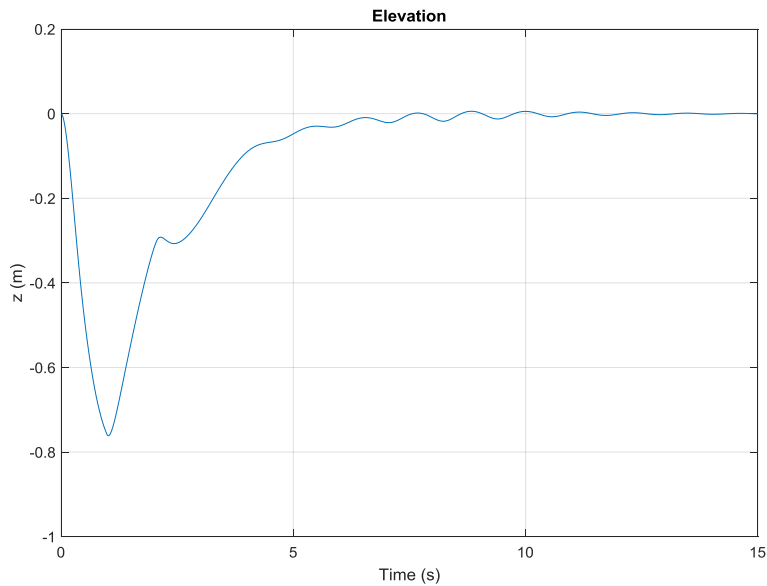
$$\dot{\hat{w}} = \beta (\Gamma^T(x)x - v |x| \hat{w})$$



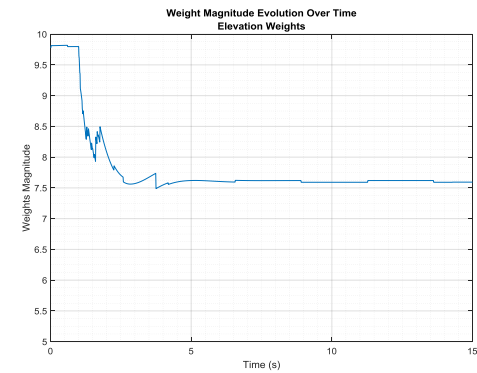
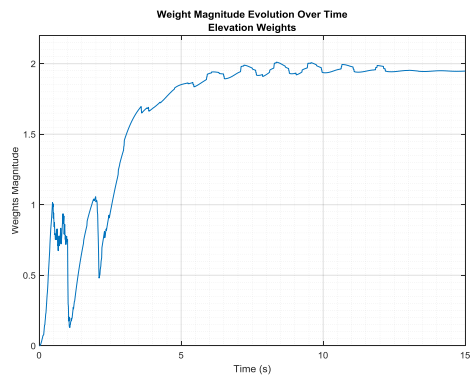
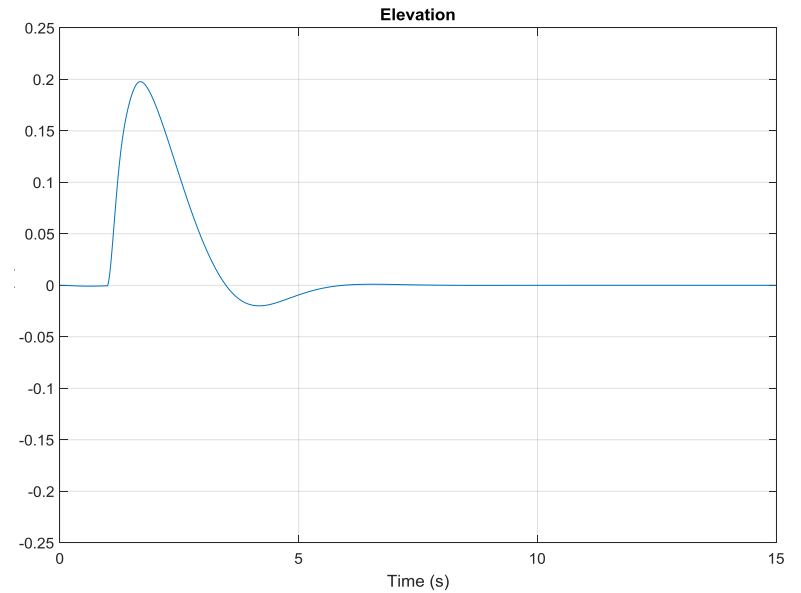
ADAPTIVE CONTROL

SIMULATION CASE 1 – MASS REDUCTION

Initial weights = 0; $b=100$, $n=0.0011$



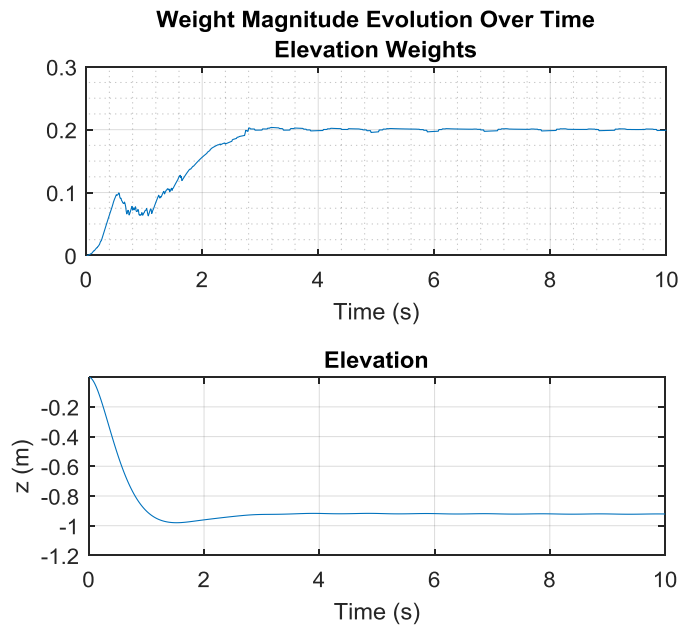
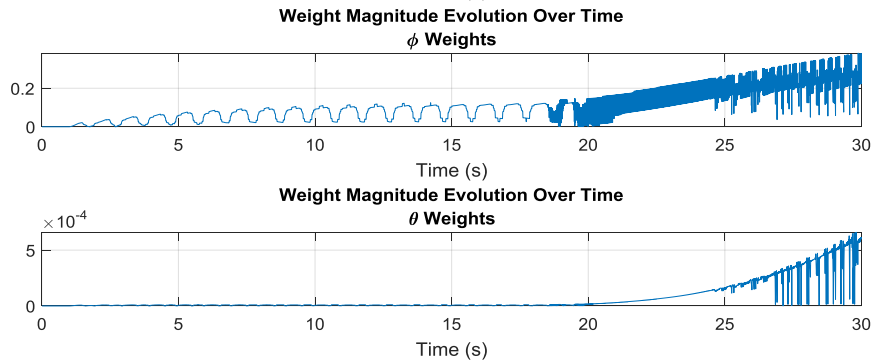
Initial weights = pre-trained; $b=100$, $n=0.0011$



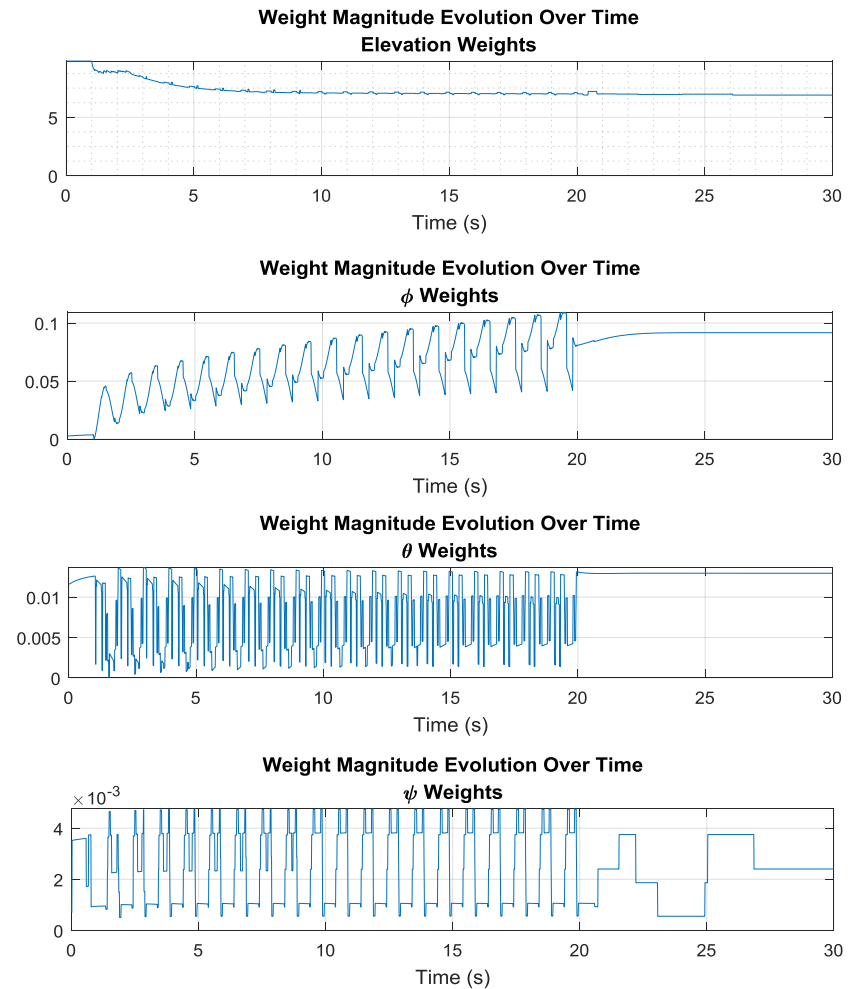
ADAPTIVE CONTROL

SIMULATION CASES 2 AND 3

Simulated Wind

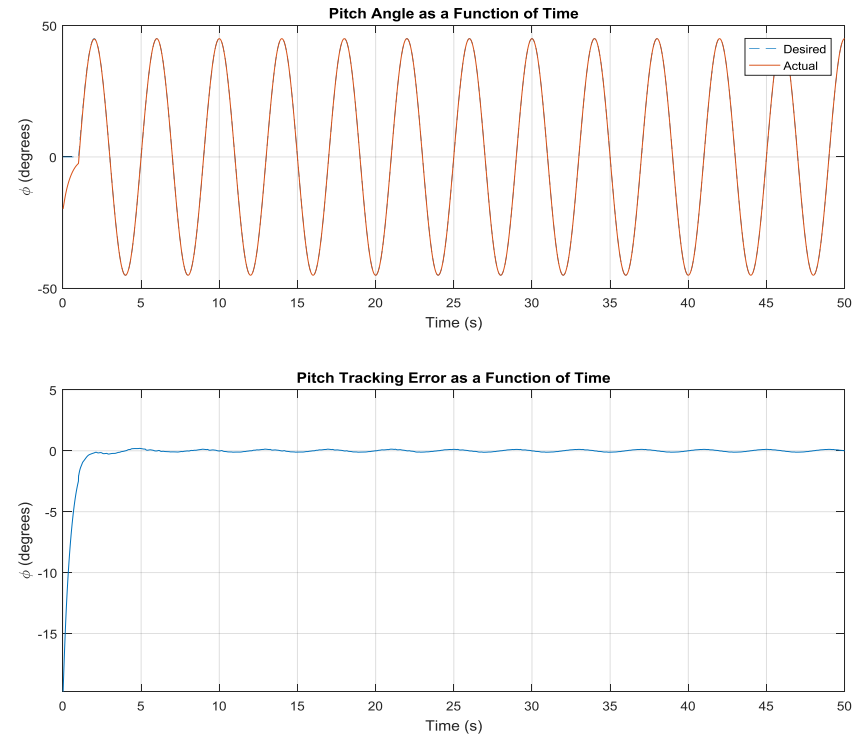
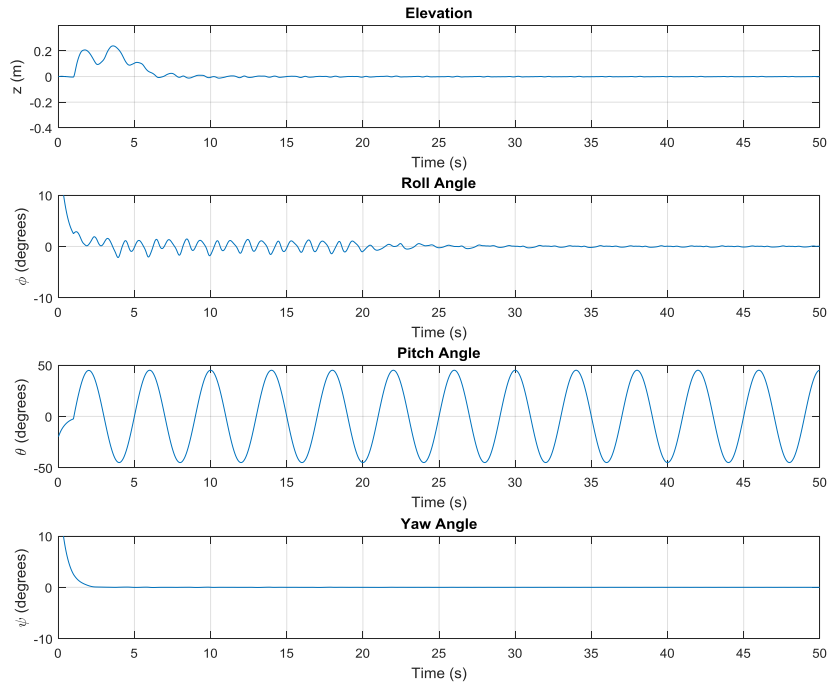


Mass Reduction + Simulated Wind Pre-trained Initial Weights



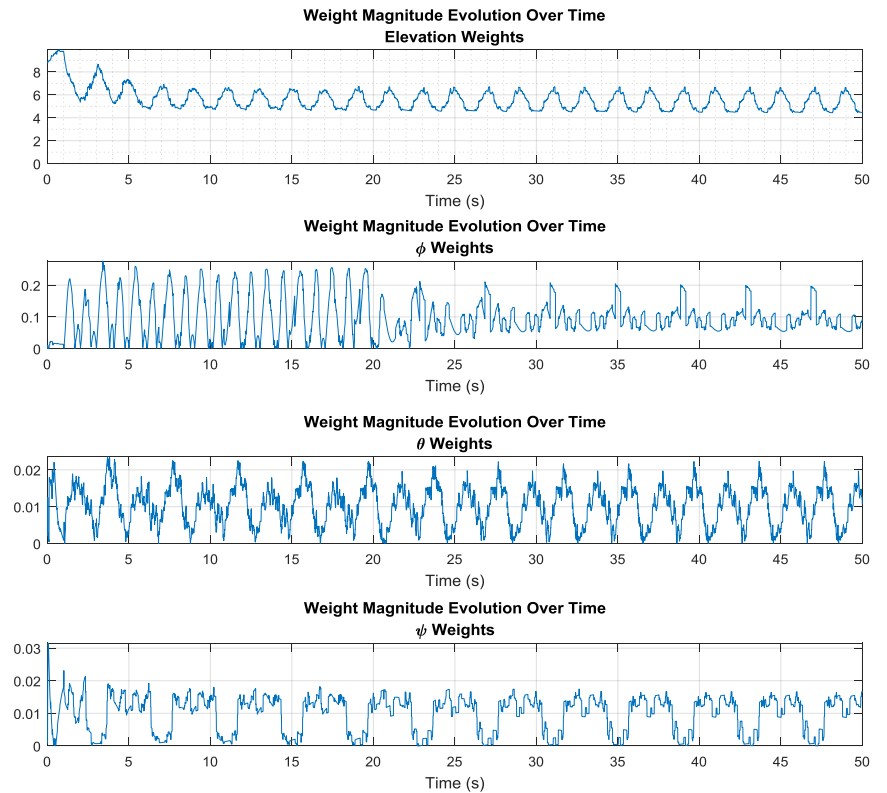
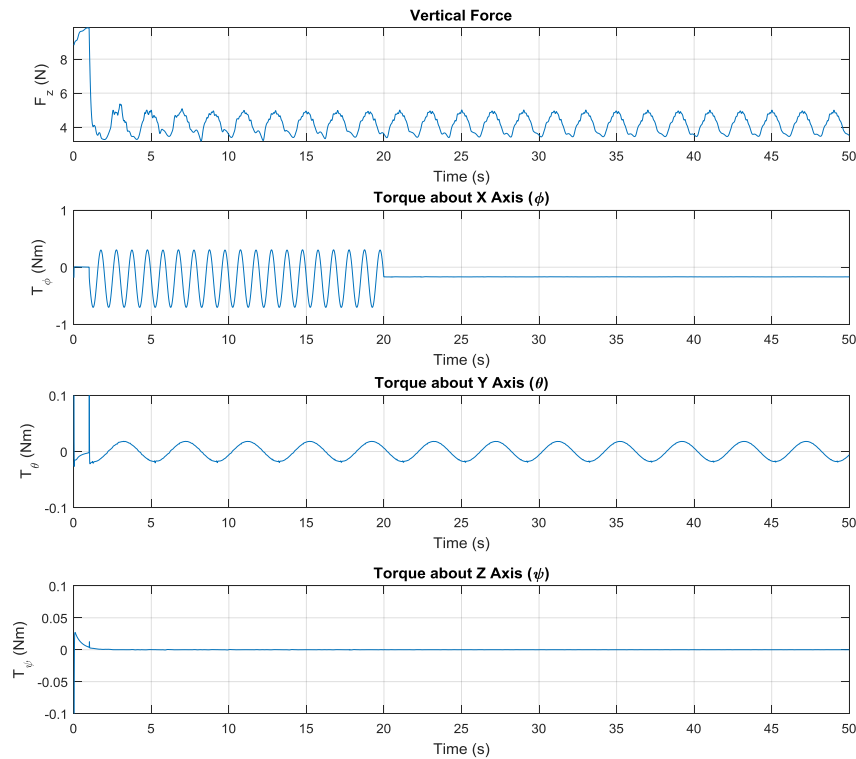
ADAPTIVE CONTROL

SIMULATION CASE 4 – TRAJECTORY WITH MASS REDUCTION AND SIMULATED WIND



ADAPTIVE CONTROL

SIMULATION CASE 4 – TRAJECTORY WITH MASS REDUCTION AND SIMULATED WIND



CONCLUSIONS

Intelligent controls were designed and tested on a simulated quadrotor

Neural network control with static training only

- CMAC neural network offered better training results than MLP
- Quadrotor responded well to sinusoidal trajectories

Adaptive neural network control

- Better to start with pre-trained weights
- Quadrotor adapted to different disturbances while tracking a trajectory input
- A lot of trial and error required for adequate selection of β and v

REFERENCES

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C. Nicol, C. Macnab and A. Ramiriz-Serrano, "Robust Adaptive Control of a Quadrotor Helicopter," Elsevier Mechatronics, 2011.

S. Ge, C. Hang, T. Lee and T. Zhang, Stable Adaptive Neural Network Control, Norwell: Kluwer Academic Publishers, 2002.